Chapter 10 — Collections

Almost every real program holds *groups* of things: the items in a cart, the users online, the word counts of a document, the jobs waiting to run. Arrays (Chapter 4) have a fixed length and almost no behaviour. The **Java Collections Framework** (in `java.util`) gives you resizable lists, duplicate-free sets, key→value maps, queues (first-in, first-out waiting lines), stacks (last-in, first-out piles) and priority queues — all sharing a small set of interfaces (contracts that list the methods a type must offer) so you can swap one implementation (a concrete class that fulfils the contract) for another without rewriting code.

This chapter teaches what each collection is, **how it works inside** (so the big-O costs — how an operation's running time grows as the collection gets bigger, explained in §10.12 — make sense rather than being memorised), which one to pick, and the traps that cause real bugs. It leans on generics (`List<String>`, `? extends T`) from Chapter 9 and on `equals`/`hashCode` from §6.13. Lambdas such as `n -> n % 2 == 0` appear in a few examples; they are explained fully in Chapter 11 — for now read them as "a small function".

What you'll learn

1. The collection hierarchy: `Iterable`, `Collection`, `List`, `Set`, `Queue`, `Deque`, and why `Map` stands apart.
2. `ArrayList` vs `LinkedList` — internals, costs and the classic traps.
3. `HashSet`, `LinkedHashSet`, `TreeSet` and navigation queries.
4. The `Map` API (`merge`, `computeIfAbsent`, …), `LinkedHashMap` (LRU cache — a cache that throws out the Least Recently Used entry when full; see System Design Fundamentals), `TreeMap`.
5. `HashMap` **internals**: hashing, buckets, load factor, resizing, treeification — with a measured collision demo.
6. `Queue`, `Deque` (`ArrayDeque` as stack and queue) and `PriorityQueue` (a heap — a tree-shaped structure that always keeps the smallest item on top; unrelated to the JVM's memory heap).
7. Iterators, fail-fast behaviour and safe removal.
8. `Comparable` vs `Comparator` and comparator composition.
9. The `Collections` utility class, unmodifiable views vs immutable copies (`List.of`, `copyOf`).
10. Sequenced collections (Java 21).
11. A big-O table, a measured performance demo, and a decision guide for choosing a collection.

All runnable code is in `code/` (package `ch10.*`), compiled with `javac --release 25` and run on JDK 26.

> **Exam depth.** `List.of` vs `Arrays.asList` puzzles, `remove(int)` vs `remove(Object)`, `TreeSet`/`TreeMap` ordering questions, `Comparator` chaining and generics wildcards are drilled in OCP Chapter 9 — Collections and generics (see its *Gotchas & exam traps* section). This chapter focuses on understanding and real use.

10.1 The Collections Framework and its hierarchy

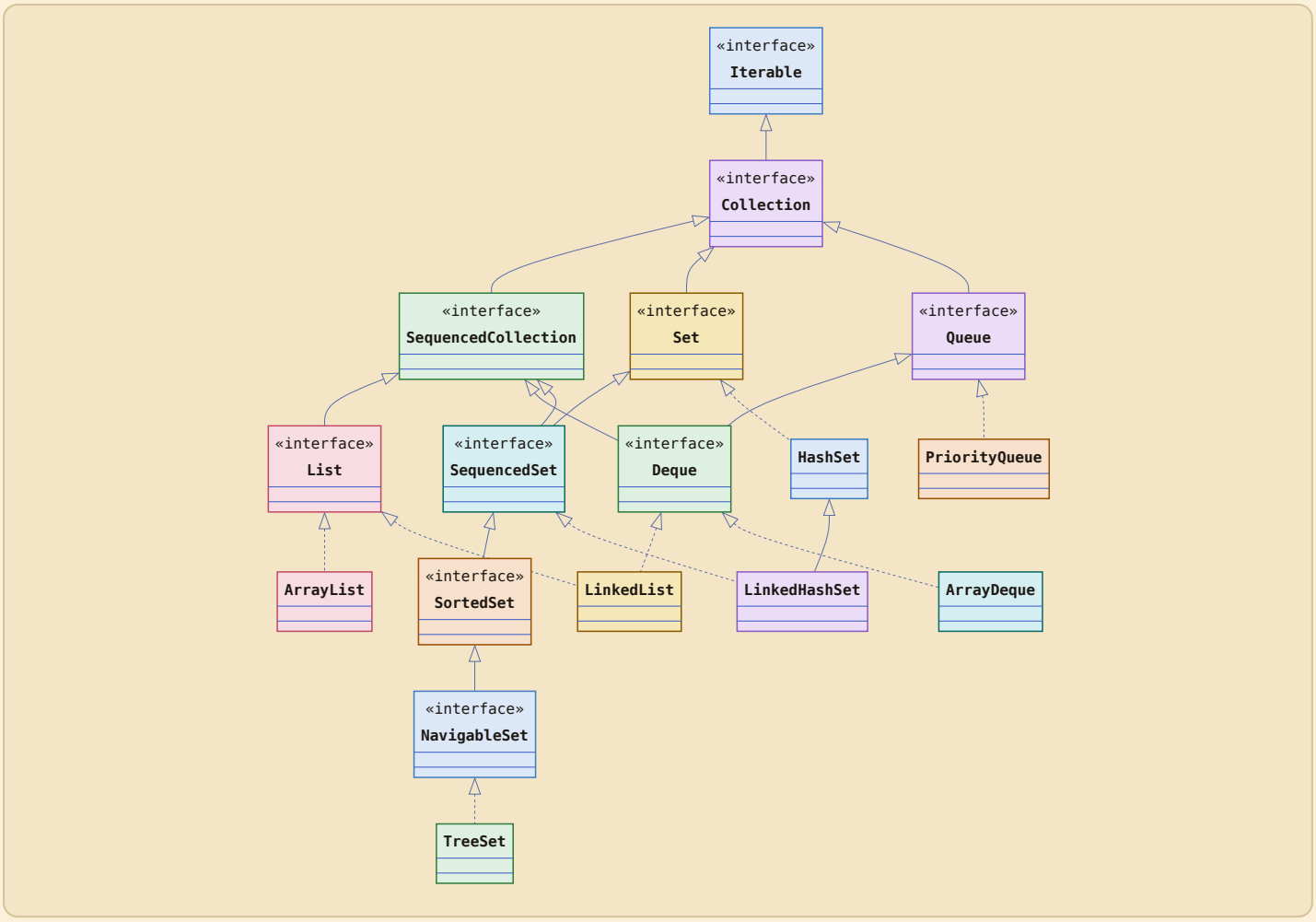
What it is

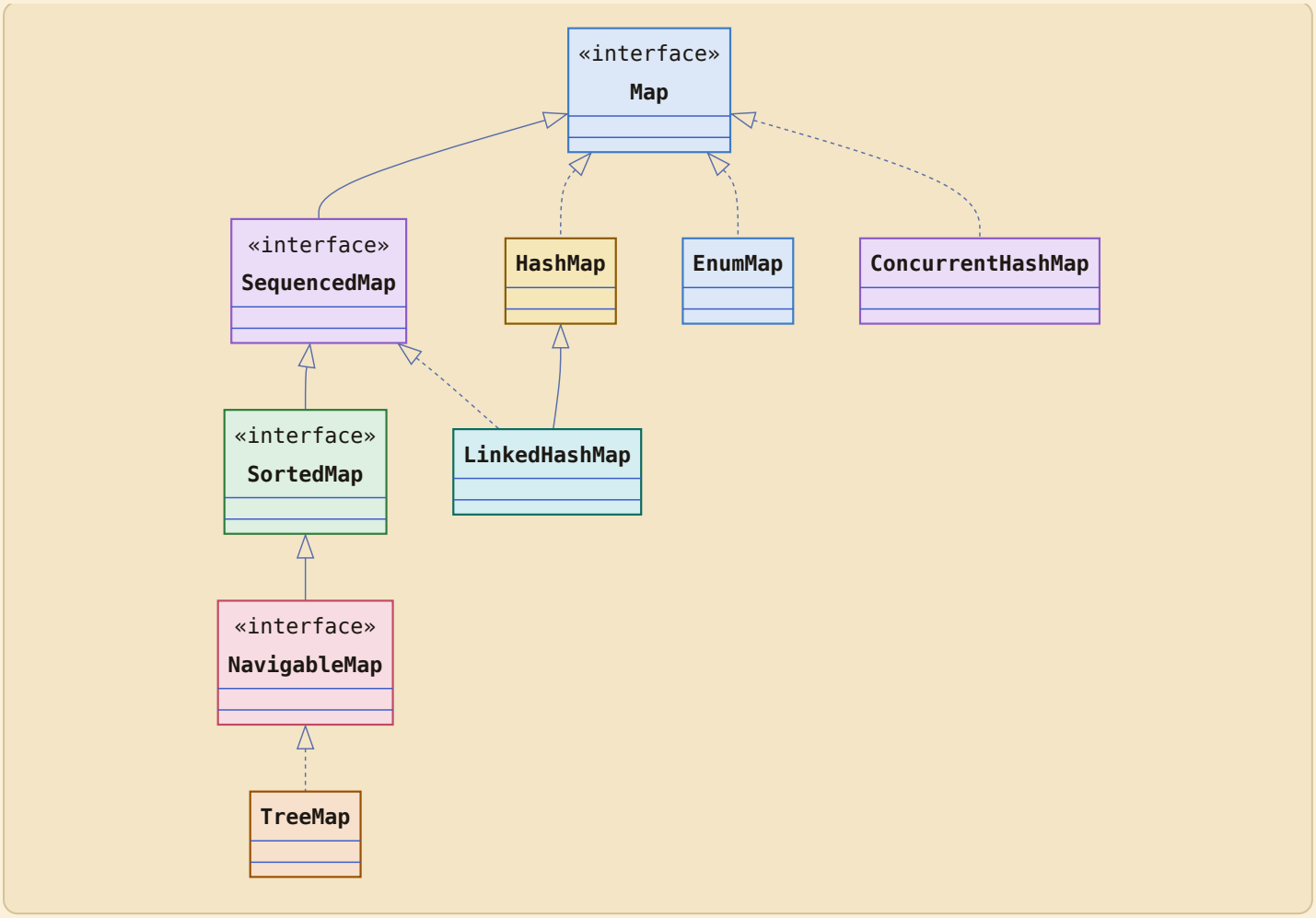
A **collection** is an object that groups other objects (its **elements**). The **Collections Framework** is three things:

1. **Interfaces** — abstract contracts such as `List` ("ordered, indexable, duplicates allowed") and `Set` ("no duplicates").
2. **Implementations** — concrete classes such as `ArrayList` and `HashSet` that fulfil those contracts with a particular data structure (a way of organising data in memory).
3. **Algorithms** — reusable static methods (sorting, searching, shuffling) in `Collections` and `Arrays`.

Analogy: the interface is the job description, the implementation the person hired — you can hire someone else without rewriting the description.

How it is organised





Interface	Promise	Main implementations
`Iterable<T>`	Can be looped over with for-each	every collection
`Collection<E>`	A group of elements: `add`, `remove`, `contains`, `size`, `iterator`, `stream`	(root of the element-based types)
`List<E>`	Ordered by index, duplicates allowed	`ArrayList`, `LinkedList`
`Set<E>`	No duplicates (by `equals`)	`HashSet`, `LinkedHashSet`, `TreeSet`, `EnumSet`
`Queue<E>`	Elements wait to be processed, usually FIFO (first in, first out)	`ArrayDeque`, `LinkedList`, `PriorityQueue`
`Deque<E>`	"Double-ended queue": add/remove at both ends; also a stack	`ArrayDeque`, `LinkedList`
`Map<K,V>`	Unique keys, each mapped to one value	`HashMap`, `LinkedHashMap`, `TreeMap`, `EnumMap`
`SequencedCollection` / `SequencedMap` (21)	Has a defined first and last element	`List`, `Deque`, `LinkedHashSet`, `TreeSet`, `LinkedHashMap`, `TreeMap` (§10.11)

Key terms used throughout:

- **Ordered / encounter order** — iteration visits elements in a defined sequence (index order, insertion order or sorted order).
- **Sorted** — ordered by comparing elements (natural order — the type's own built-in ordering — or a `Comparator`, §10.8).

- **Structural modification** — a change in size (add/remove), as opposed to replacing an element with `set`.
- **Optional operation** — an interface method an implementation may reject with `UnsupportedOperationException` (e.g. `add` on an unmodifiable list).

`Map` is not a `Collection`. A map stores pairs, so `add(E)` makes no sense for it. Instead it offers three **views** (live windows onto the map's data — changes show through both ways) that *are* collections: `keySet()`, `values()` and `entrySet()`.

Program to the interface

```
// code/ch10/hierarchy/HierarchyDemo.java (key parts)
Collection<String> langs = new ArrayList<>();          // declared type = interface
langs.add("Java"); langs.add("Go"); langs.add("Java"); // lists allow duplicates
langs.addAll(List.of("Rust", "Kotlin"));
langs.remove("Java");                                  // removes the FIRST match only
langs.removeIf(s -> s.length() <= 2);                  // bulk remove by condition (Java 8)

Collection<String> unique = new HashSet<>(List.of("Java", "Go", "Rust"));
System.out.println("add duplicate to set -> " + unique.add("Java"));   // add returns false

String[] arr = langs.toArray(String[]::new);           // typed array (Java 11 form)
Map<String, Integer> ages = new TreeMap<>(Map.of("ana", 31, "bo", 25));
System.out.println("keys=" + ages.keySet() + " values=" + ages.values() + " entries=" + ages.entrySet());
```

Output:

```
list-backed: [Java, Go, Java, Rust, Kotlin] size=5
contains Go? true
after removes: [Java, Rust, Kotlin]
add duplicate to set -> false
array: [Java, Rust, Kotlin] isEmpty=false
keys=[ana, bo] values=[31, 25] entries=[ana=31, bo=25]
Map is a Collection? false
```

Declaring variables and parameters as `List`, `Set`, `Map` (not `ArrayList`…) means the method works with any implementation and you can change your mind in one place. Use the concrete type only when you need its extra methods (e.g. `TreeMap.floorKey`, `LinkedList.addFirst` before Java 21).

Legacy classes you will still meet

`Vector` (synchronized `ArrayList`), `Stack` (extends `Vector`), `Hashtable` (synchronized map, no nulls) and `Enumeration` date from Java 1.0. They still work but every method takes a lock (only one thread at a time may run it, which costs time), and `Stack` inherits index methods a stack should not have. Use `ArrayList`, `ArrayDeque`, `HashMap`, and — when threads (independent paths of execution running at the same time) share the collection — the `java.util.concurrent` classes (Chapter 14).

Self-check

1. Which interface lets a class be used in a for-each loop? → `Iterable`.
2. Is `Map` a `Collection`? → No; it provides collection *views* (`keySet`, `values`, `entrySet`).
3. Why declare `List<String> xs = new ArrayList<>()` rather than `ArrayList<String>`? → Code depends only on the contract, so the implementation can change without touching callers.

TL;DR

Interfaces (`List`, `Set`, `Queue`, `Deque`, `Map`) describe behaviour; classes (`ArrayList`, `HashSet`, `ArrayDeque`, `HashMap`, `TreeMap`) pick the data structure. Program to the interface. Avoid `Vector`, `Stack`, `Hashtable`.

10.2 Lists: `ArrayList` and `LinkedList`

What it is

A `List` is an ordered sequence you can index into (`get(i)`), like a numbered shopping list. It allows duplicates and (for most implementations) `null`.

- `ArrayList` — a resizable array. Analogy: a row of numbered lockers; jumping to locker 500 is instant, but inserting a new locker at position 0 means shifting every locker one place.
- `LinkedList` — a **doubly linked list**: each element sits in a small *node* object holding pointers (references) to the previous and next node. Analogy: a treasure hunt — each clue tells you where the next one is, so reaching clue 500 means following 500 clues.

How `ArrayList` works

1. It holds an internal `Object[] elementData` and an `int size`.
2. `new ArrayList<>()` starts with a shared empty array; the first `add` allocates capacity **10**.
3. `add(e)` puts `e` at `elementData[size++]`. When the array is full it allocates a new one about **1.5×** larger (`old + old >> 1`) and copies everything across. Because growth is geometric, the total copying is proportional to n, so `add` at the end is **amortised O(1)** (occasionally O(n), on average constant).
4. `get(i)` / `set(i, e)` are one array access: **O(1)**.
5. `add(i, e)` and `remove(i)` shift the tail with `System.arraycopy`: **O(n − i)**.
6. `contains`/`indexOf` scan linearly with `equals`: **O(n)**.
7. It never shrinks on its own; `trimToSize()` releases unused capacity. `new ArrayList<>(expectedSize)` or `ensureCapacity(n)` avoid repeated growth.

How `LinkedList` works

1. Each element is wrapped in a `Node { item, prev, next }`; the list keeps `first`, `last` and `size`.
2. Adding or removing at either end rewires two pointers: **O(1)**.
3. `get(i)` walks from whichever end is closer: **O(n)**.
4. Inserting in the middle is O(1) *only once you are already there* (via a `ListIterator`, an iterator that can move both ways and insert — §10.7); getting there is O(n).
5. Each node is a separate heap object (about 24 bytes of overhead each on a typical 64-bit JVM with compressed pointers — references squeezed into 32 bits to save memory), and nodes are scattered in memory, which defeats the CPU cache (a small, very fast memory beside the processor that works best when data sits side by side).

`LinkedList` also implements `Deque`, so it can act as a queue or stack — but `ArrayDeque` does that job faster (§10.6). In practice `ArrayList` **is the right default for almost every list**; the measured demo in §10.12 shows why.

Code

```java
// code/ch10/lists/ListDemo.java (key parts)
List<String> todo = new ArrayList<>(List.of("write", "test", "ship"));
todo.add(1, "review");                  // insert at index 1, shifts the rest right
todo.set(0, "design");                  // replace; returns the old value

List<Integer> nums = new ArrayList<>(List.of(10, 20, 30, 1));
nums.remove(1);                         // int argument -> remove(int index): removes 20
nums.remove(Integer.valueOf(1));        // object argument -> remove(Object): removes the value 1

List<String> firstTwo = todo.subList(0, 2);
firstTwo.clear();                       // subList is a VIEW: clears those elements from todo

String[] arr = {"a", "b", "c"};
List<String> view = Arrays.asList(arr); // fixed-size view backed by the array
view.set(0, "z");                        // writes through to arr
view.add("d");                           // UnsupportedOperationException

LinkedList<String> ll = new LinkedList<>(List.of("b", "c"));
ll.addFirst("a"); ll.addLast("d");

List<String> names = new ArrayList<>(List.of("mia", "Al", "zoe", "Bob", "ann"));
names.sort(null);                        // null comparator = natural order
names.sort(String.CASE_INSENSITIVE_ORDER);
```

Output:

```
[design, review, test, ship] get(2)=test indexOf(ship)=3
nums=[10, 30]
after subList.clear(): [test, ship]
array now [z, b, c]
asList.add -> UnsupportedOperationException
linked=[a, b, c, d] removeFirst=a peekLast=d
natural (uppercase first): [Al, Bob, ann, mia, zoe]
case-insensitive: [Al, ann, Bob, mia, zoe]
ArrayList equals LinkedList? true
```

Notes on the output:

- `remove(1)` on a `List<Integer>` picks the **`remove(int index)`** overload because `1` is an `int` and no boxing (wrapping a primitive in its object type, `int` → `Integer`) is needed (overload resolution, §4.3). Box explicitly to remove by value.
- Natural `String` order compares UTF-16 code units (the 16-bit numbers Java stores for each character), so every uppercase letter sorts before every lowercase one.
- `List.equals` is defined by the interface: same elements in the same order, regardless of implementation.

Trade-offs

Operation	ArrayList	LinkedList
`get(i)` / `set(i, e)`	**O(1)**	O(n)
`add(e)` at end	amortised **O(1)**	O(1)
`add(0, e)` / `remove(0)`	O(n) (shift)	**O(1)**
insert/remove in middle	O(n) shift	O(n) to find + O(1) to link
`contains` / `indexOf`	O(n)	O(n)
Memory per element	one reference (+ spare capacity)	node object ≈ 24 B + reference
Cache friendliness	high (contiguous — side by side in memory)	low (pointer chasing — hopping from node to node)

When not to use them

- Frequent "is x in here?" checks → use a `Set` (O(1) or O(log n)) instead of `list.contains` (O(n)).
- Queue/stack work (add/remove at the ends) → `ArrayDeque`, not `ArrayList.remove(0)`.
- Fixed-size numeric data in hot loops → a primitive array (`int[]`) avoids boxing each element into an `Integer`.

Common mistakes

- `remove(int)` vs `remove(Object)` on a `List<Integer>` (above).
- Holding a `subList` and then structurally modifying the parent: the sublist becomes invalid and throws `ConcurrentModificationException` on next use.
- Expecting `Arrays.asList(...)` to be growable. Wrap it: `new ArrayList<>(Arrays.asList(...))`.
- `Arrays.asList(intArray)` gives a `List<int[]>` with **one** element — generics can't hold primitives.
- Calling `list.get(i)` in a loop over a `LinkedList` — O(n²) overall; use for-each or an iterator.

In the interview

> **30-second answer:** "ArrayList is a growable array: O(1) random access, amortised O(1) append, O(n) inserts or removes in the middle because it shifts elements. LinkedList is a doubly linked list: O(1) at the ends but O(n) to reach an index, more memory per element and poor cache locality. I default to ArrayList, and use ArrayDeque when I need a queue or stack."

- *Follow-up: How does ArrayList grow?* → Starts at 10 on first add, then about 1.5× each time with an array copy; geometric growth makes append amortised O(1).
- *Follow-up: When would LinkedList actually win?* → Many inserts/removals at positions you are already at via a `ListIterator`, in a very long list — rare in practice; measure first.
- *Follow-up: Difference between `List.of` and `Arrays.asList`?* → `List.of` is fully unmodifiable and rejects nulls; `Arrays.asList` is a fixed-size, writable view of an array that allows nulls (§10.10).

Self-check

1. What is the cost of `list.add(0, x)` on an `ArrayList` of n elements? → O(n) — every element shifts.
2. `List<Integer> l = new ArrayList<>(List.of(5, 6, 7)); l.remove(2);` — what's left? → `[5, 6]` (index 2 removed).

TL;DR

`ArrayList` = array + 1.5× growth: fast index and append, slow front/middle edits. `LinkedList` = linked nodes: fast ends, slow indexing, heavy. Default to `ArrayList`.

10.3 Sets: `HashSet`, `LinkedHashSet`, `TreeSet`

What it is

A `Set` holds each element at most once — "at most once" meaning no two elements are `equals`. Analogy: a guest list; writing a name twice doesn't add a second guest.

Class	Built on	Order	`add` / `contains` / `remove`	Nulls
`HashSet`	a `HashMap` (elements are the keys)	none guaranteed	**O(1)** average	one `null` allowed
`LinkedHashSet`	a `LinkedHashMap`	insertion order	O(1) average	one `null`
`TreeSet`	a `TreeMap` (red-black tree, see below)	sorted	**O(log n)**	no `null` with natural order
`EnumSet`	a bit vector (one bit per enum constant)	enum declaration order	O(1), extremely fast	no `null` (§6.9)

How it works

- `HashSet` literally stores each element as a key in an internal `HashMap`, with a shared dummy object as the value. So everything in §10.5 applies: it uses `hashCode()` to find a bucket and `equals()` to confirm a match.
- `LinkedHashSet` adds a doubly linked list through the entries, remembering insertion order (re-adding an existing element does not move it).
- `TreeSet` keeps elements in a **red-black tree**, a self-balancing binary search tree (each node's left subtree holds smaller values and its right subtree larger ones) whose height stays O(log n). It never calls `equals` or `hashCode`: two elements are "the same" when `compareTo`/`compare` returns 0.

Code

```
// code/ch10/sets/SetDemo.java (key parts)
List<String> visits = List.of("home", "cart", "home", "search", "cart", "about");
new HashSet<>(visits);  new LinkedHashSet<>(visits);  new TreeSet<>(visits);

Set<String> union = new TreeSet<>(a); union.addAll(b);      // A ∪ B
Set<String> inter = new TreeSet<>(a); inter.retainAll(b);   // A ∩ B
Set<String> diff  = new TreeSet<>(a); diff.removeAll(b);    // A − B

TreeSet<Integer> prices = new TreeSet<>(List.of(5, 12, 20, 35, 50));
prices.floor(19); prices.ceiling(19); prices.lower(20); prices.higher(20);
prices.headSet(20); prices.tailSet(20); prices.subSet(10, 40);

List<String> key = new ArrayList<>(List.of("x"));
Set<List<String>> set = new HashSet<>(); set.add(key);
key.add("y");                                               // mutates a key already in the set
```

Output:

```
HashSet       [search, about, cart, home]  (order not guaranteed)
LinkedHashSet [home, cart, search, about]  (insertion order)
TreeSet       [about, cart, home, search]  (sorted)
union=[docker, git, java, sql] intersection=[git] difference=[java, sql]
floor(19)=12 ceiling(19)=20 lower(20)=12 higher(20)=35
headSet(20)=[5, 12] tailSet(20)=[20, 35, 50] subSet(10,40)=[12, 20, 35]
first=5 pollLast=50 descending=[35, 20, 12, 5]
contains mutated key? false, size=1
HashSet null ok=true, TreeSet.add(null) -> NPE
```

Navigation methods (`NavigableSet`): `floor(x)` = greatest element ≤ x, `ceiling(x)` = least ≥ x, `lower`/`higher` = strictly < / >. `headSet(x)` is exclusive of x, `tailSet(x)` inclusive, `subSet(from, to)` is `[from, to)`. These range views are *live*: changes show through both ways.

The mutated-key line is the most important bug in this section: after `key.add("y")` the list's `hashCode` changed, so `contains` looks in a different bucket and reports `false` even though the object is in the set. **Never mutate an object while it is a set element or map key** — use immutable keys (strings, records of immutable fields).

Common mistakes

- Putting objects whose class overrides `equals` but not `hashCode` into a `HashSet` → duplicates appear (§6.13).
- A `TreeSet` comparator that returns 0 for different elements silently drops them (§10.8).
- Relying on `HashSet` print order in tests — it can change between JDK versions or after a resize.

Real-world examples

Unique visitors per day (`HashSet<UserId>`), de-duplicating while keeping first-seen order (`LinkedHashSet`), a leaderboard or an index of time-stamped events that needs "next event after t" (`TreeSet.ceiling`), feature flags (on/off switches for features; `EnumSet`).

Self-check

1. Which set gives insertion order? → `LinkedHashSet`.
2. What does `TreeSet` use to detect duplicates? → `compareTo`/`compare` returning 0, not `equals`.
3. `new TreeSet<>(List.of(1, 5, 9)).floor(4)`? → `1`.

TL;DR

`HashSet` for fast membership, `LinkedHashSet` to also keep insertion order, `TreeSet` for sorted order and range/nearest queries (O(log n)). Keys must not change while inside.

10.4 Maps: `HashMap`, `LinkedHashMap`, `TreeMap`

What it is

A `Map<K,V>` associates each unique **key** with one **value** — a dictionary: look up a word (key), get its definition (value). Putting a second value under an existing key *replaces* the old value.

The everyday API

Method	Does	Returns
`put(k, v)`	insert or replace	previous value or `null`
`get(k)` / `getOrDefault(k, d)`	look up	value, or `null` / `d` if absent
`containsKey(k)` / `containsValue(v)`	test	`boolean` (`containsValue` is O(n))
`remove(k)`	delete	removed value or `null`
`putIfAbsent(k, v)`	insert only if missing (or mapped to `null`)	existing value or `null`
`merge(k, v, fn)`	if absent put `v`, else put `fn(old, v)`; `null` result removes	new value
`compute(k, fn)`	`fn(k, oldOrNull)`; `null` result removes	new value
`computeIfAbsent(k, fn)`	if missing, compute and store `fn(k)`	current value
`computeIfPresent(k, fn)`	if present, recompute; `null` removes	new value
`forEach((k, v) -> …)`, `replaceAll((k, v) -> …)`	iterate / transform all values	—
`keySet()`, `values()`, `entrySet()`	live views	collections

The `merge`/`compute*` methods arrived in Java 8 and replace most "get, check for null, put" boilerplate (repetitive code).

Code

```java
// code/ch10/maps/MapDemo.java (key parts)
Map<String, Integer> stock = new HashMap<>();
stock.put("apple", 5);
Integer old = stock.put("apple", 7);                    // replaces, returns 5
stock.putIfAbsent("pear", 3);
stock.merge("apple", 10, Integer::sum);                 // 7 + 10
stock.merge("plum", 1, Integer::sum);                   // absent -> puts 1
stock.compute("pear", (k, v) -> v == null ? 1 : v * 2);
stock.computeIfPresent("plum", (k, v) -> null);         // null result REMOVES plum

Map<String, Integer> counts = new TreeMap<>();          // word count idiom
for (String w : "to be or not to be".split(" ")) counts.merge(w, 1, Integer::sum);

Map<Character, List<String>> byInitial = new TreeMap<>(); // multimap idiom
for (String n : List.of("ana", "bo", "al", "cy", "ben"))
    byInitial.computeIfAbsent(n.charAt(0), c -> new ArrayList<>()).add(n);

for (Map.Entry<String, Integer> e : counts.entrySet())  // iterate entries, not keys + get
    if (e.getValue() > 1) System.out.print(e.getKey() + "x" + e.getValue() + " ");

TreeMap<Integer, String> bracket = new TreeMap<>(Map.of(0, "10%", 11_000, "12%", 44_725, "22%", 95_375, "24%"));
bracket.floorEntry(50_000).getValue();                  // bracket whose start <= 50,000

Map<String, String> lru = new LinkedHashMap<>(16, 0.75f, true) {   // true = access order
    @Override protected boolean removeEldestEntry(Map.Entry<String, String> eldest) { return size() > 3; }
};
lru.put("a", "A"); lru.put("b", "B"); lru.put("c", "C");
lru.get("a");       // a becomes most recently used
lru.put("d", "D");  // evicts b
```

Output:

```
old=5 get=7 missing=null default=0
stock={apple=17, pear=6}
counts={be=2, not=1, or=1, to=2}
byInitial={a=[ana, al], b=[bo, ben], c=[cy]}
bex2 tox2
income 50000 -> 22%, first bracket above 50000 starts at 95375, headMap(44725)={0=10%, 11000=12%}
LRU keys=[c, a, d]
HashMap null key -> 0
```

(The brackets are illustrative numbers in the style of a progressive income-tax table, not current law.)

The three main implementations

	HashMap	LinkedHashMap	TreeMap
Structure	array of buckets (§10.5)	`HashMap` + doubly linked list of entries	red-black tree
Iteration order	unspecified	insertion order (or access order)	sorted by key
`get` / `put` / `remove`	**O(1)** average	O(1) average	**O(log n)**
`null` key	one allowed	one allowed	not with natural ordering
Extras	—	`removeEldestEntry` hook → LRU cache	`floorKey`, `ceilingEntry`, `headMap`, `tailMap`, `subMap`, `firstKey`, `descendingMap`

Other maps worth knowing: **`EnumMap`** (enum keys, backed by an array indexed by `ordinal()` — very fast), **`IdentityHashMap`** (compares keys with `==`, used for object-graph algorithms like serialization — turning objects into bytes), **`WeakHashMap`** (keys held weakly — through references that don't stop garbage collection — so an entry disappears once nothing else references its key — for caches of metadata), **`ConcurrentHashMap`** (thread-safe — safe for many threads at once — no null keys/values — Chapter 14; a simplified build is in LLD §13.7).

Common mistakes

- `if (map.get(k) != null)` to test presence — wrong if `null` values are allowed; use `containsKey`.
- `for (K k : map.keySet()) use(map.get(k))` — two lookups per entry; iterate `entrySet()` or use `map.forEach`.
- `map.get(k) + 1` when `k` is absent → `NullPointerException` from unboxing `null`. Use `merge` or `getOrDefault`.
- In `computeIfAbsent`, the mapping function must not modify the same map (for `HashMap` this throws `ConcurrentModificationException` since Java 9). This bites recursive memoisation (caching the results of earlier calls) — see §11.8.

Real-world examples

HTTP headers (key–value metadata sent with each web request) and configuration (`Map<String,String>`), caches keyed by ID, word or event counts (`merge`), grouping orders by customer (`computeIfAbsent`), an LRU cache for thumbnails (`LinkedHashMap`), time-series (data points indexed by time) lookups and rate tables (`TreeMap.floorEntry`).

Self-check

1. What does `map.put(k, v)` return when `k` was already present? → The previous value.
2. How do you count occurrences in one line per element? → `counts.merge(word, 1, Integer::sum)`.
3. Which map gives "the entry with the largest key ≤ x"? → `TreeMap.floorEntry(x)`.
4. What does returning `null` from a `compute` function do? → Removes the key.

TL;DR

Map = unique keys → values. `HashMap` default, `LinkedHashMap` for predictable order or LRU, `TreeMap` for sorted keys and range queries. Use `merge`, `computeIfAbsent`, `getOrDefault` instead of null-check boilerplate.

10.5 `HashMap` internals

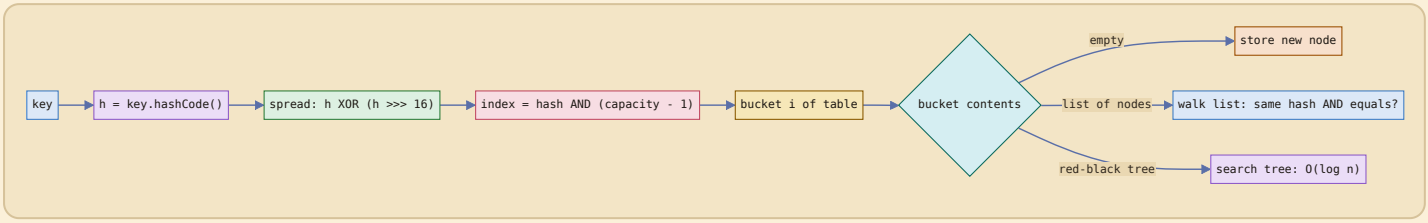
What it is

`HashMap` is a **hash table**: an array of **buckets** (also called *bins*), where a key's **hash code** (an `int` computed from the key's contents by `hashCode()`) decides which bucket it goes into. Analogy: a coat check with 16 numbered hooks. The attendant computes a number from your ticket, hangs your coat on that hook, and later goes straight to that hook — no searching through every coat. If two coats land on the same hook, they hang one behind the other, and the attendant checks each ticket on that hook.

Why it matters

Average **O(1)** `get`/`put` makes `HashMap`/`HashSet` the most-used collections — and a favourite interview topic. The internals also explain real bugs: slow maps from bad `hashCode`, lost keys from mutation, surprising iteration order, wasted resizes.

How it works (JDK 8 and later)



1. **Hash.** `put(key, value)` calls `key.hashCode()` (a 32-bit `int`). A `null` key is given hash 0 (so it lives in bucket 0).
2. **Spread.** `HashMap.hash()` XORs (bitwise exclusive-or: a result bit is 1 where the two input bits differ) the high 16 bits into the low 16: `h ^ (h >>> 16)`. The next step only looks at the low bits, and without spreading, keys whose hash codes differ only in high bits would all collide.
3. **Index.** The table length `n` is always a **power of two** (default **16**), so `hash & (n − 1)` equals `hash mod n` (the remainder after dividing by n) but is a single cheap AND (bitwise and: keeps only bits set in both) instruction.
4. **Store.** Each bucket holds a chain of `Node{hash, key, value, next}`. If the bucket is empty, the node goes straight in. Otherwise the map walks the chain: a node matches when **the hashes are equal and** (`==` or `equals`). A match gets its value replaced (and `put` returns the old value); no match → a new node is appended at the tail.
5. **Load factor and resize.** The **load factor** (default **0.75**) is the fullness at which the table grows. The **threshold** = capacity × load factor (16 × 0.75 = **12**). When `size` exceeds the threshold, the table **doubles** (16 → 32 → 64 …) and every node is redistributed. Because capacity is a power of two, each node either stays at index `i` or moves to `i + oldCapacity`, decided by one bit of its hash (`hash & oldCapacity`), so the split is cheap and keeps chain order.
6. **Treeification (Java 8).** If a single bucket's chain reaches **8** nodes (`TREEIFY_THRESHOLD`) and another is added, the chain is converted into a **red-black tree** of `TreeNode`s, turning that bucket's worst case from O(n) into **O(log n)**. But if the whole table is smaller than **64** buckets (`MIN_TREEIFY_CAPACITY`), the map **resizes instead**, because a small table crowded in one bucket is better fixed by spreading keys out. When a resize split leaves a tree bucket with **≤ 6** nodes (`UNTREEIFY_THRESHOLD`), it turns back into a list. With a decent hash function and load factor 0.75, a bucket of 8 is astronomically unlikely (the JDK source comment estimates about 6 in 100 million), so trees are a defence against bad or malicious hash codes, not a normal case.

7. **Inside a tree bucket**, nodes are ordered by hash; for equal hashes, by `compareTo` **if the keys are `Comparable` and of the same class**. Otherwise the tree cannot tell them apart and a lookup may have to search both subtrees — which is why the non-`Comparable` case in the demo below stays slow.
8. **Get** repeats steps 1–3 and searches only that one bucket.

The constants, straight from `java/util/HashMap.java` in JDK 26:

Constant	Value	Meaning
`DEFAULT_INITIAL_CAPACITY`	16	buckets allocated on first `put` (the table is created lazily)
`DEFAULT_LOAD_FACTOR`	0.75	resize when size > capacity × 0.75
`TREEIFY_THRESHOLD`	8	chain length that triggers list → tree
`UNTREEIFY_THRESHOLD`	6	tree → list when a resize split leaves this few
`MIN_TREEIFY_CAPACITY`	64	below this table size, resize instead of treeifying
`MAXIMUM_CAPACITY`	2^30	table never grows beyond this

Why 0.75? It's a time/space trade-off: a lower load factor means fewer collisions but more empty buckets (memory); a higher one packs buckets tighter but lengthens chains. 0.75 keeps chains very short on average while wasting at most about a quarter of the table.

Watching it happen: a teaching-size hash map

`TinyHashMap` below implements steps 1–5 in ~60 lines (no trees, no null keys) and prints what it does. It starts with 4 buckets so we can watch two resizes.

```
// code/ch10/maps/TinyHashMap.java (key parts)
static int spread(Object key) { int h = key.hashCode(); return h ^ (h >>> 16); }

void put(K key, V value) {
    int h = spread(key), i = h & (table.length - 1);          // bucket index
    for (Node<K, V> n = table[i]; n != null; n = n.next())
        if (n.hash() == h && n.key().equals(key)) { table[i] = replace(table[i], key, value); return; }
    table[i] = new Node<>(h, key, value, table[i]);           // new node in bucket i
    if (++size > table.length * loadFactor) resize();         // threshold = capacity * 0.75
}

private void resize() {                                       // double and re-bucket every node
    Node<K, V>[] old = table;
    table = (Node<K, V>[]) new Node[old.length * 2];
    for (Node<K, V> head : old)
        for (Node<K, V> n = head; n != null; n = n.next()) {
            int i = n.hash() & (table.length - 1);            // stays at i or moves to i + oldCapacity
            table[i] = new Node<>(n.hash(), n.key(), n.value(), table[i]);
        }
}
```

Output:

```
put apple   hashCode=   93029210 spread=   93030097 index(n=4)=1
put banana  hashCode=-1396355227 spread=-1396317280 index(n=4)=0
put cherry  hashCode=-1361513063 spread=-1361552575 index(n=4)=1
put date    hashCode=    3076014 spread=    3075968 index(n=4)=0
  resize: size=4 exceeded threshold 3 -> capacity 4 -> 8
put elder   hashCode=   96592394 spread=   96593867 index(n=8)=3
put fig     hashCode=     101380 spread=     101381 index(n=8)=5
put grape   hashCode=   98615627 spread=   98616491 index(n=8)=3
  resize: size=7 exceeded threshold 6 -> capacity 8 -> 16
final table (capacity 16):
  bucket  0: date -> banana -> null
  bucket  1: cherry -> apple -> null
  bucket  5: fig -> null
  bucket 11: elder -> grape -> null
get(fig)=99 get(kiwi)=null size=7
```

Read it like this: with 4 buckets, `apple` and `cherry` collide in bucket 1. After the 4th key, size 4 > 4 × 0.75 = 3, so the table doubles. After the final resize to 16, `elder` and `grape` still share bucket 11 because their spread hashes agree in the lowest 4 bits. `put("fig", 99)` finds the existing key and replaces the value, so size stays 7. (`String.hashCode` is a fixed, documented formula — `s[0]·31^(n−1) + … + s[n−1]` — so these numbers are the same on every JVM.)

What bad hash codes cost (measured)

`HashCollisionDemo` puts 20,000 keys into a `HashMap` whose `hashCode()` returns the constant 42 — every key collides into one bucket — and times 60,000 lookups. **Timings are machine-dependent** (this run: Apple M4 Pro, JDK 26); compare the ratios.

```
// code/ch10/maps/HashCollisionDemo.java (key parts)
record CmpKey(int id) implements Comparable<CmpKey> {          // constant hash, but Comparable
    @Override public int hashCode() { return 42; }
    @Override public int compareTo(CmpKey o) { return Integer.compare(id, o.id); }
}
record PlainKey(int id) { @Override public int hashCode() { return 42; } }  // constant hash, not Comparable
record GoodKey(int id) {}                                        // record's generated hashCode
```

Output:

```
good hashCode (GoodKey)            n=20,000  3n lookups:      0.7 ms
constant hash, Comparable (tree)   n=20,000  3n lookups:      6.5 ms
constant hash, not Comparable      n=20,000  3n lookups:   1648.2 ms
```

- **Good hash:** keys spread across buckets → O(1) each.
- **Constant hash + `Comparable`:** one giant bucket, but it was treeified and ordered by `compareTo` → O(log n) each; ~10× slower than good hashing, still fast.
- **Constant hash, not `Comparable`:** the tree can't order equal-hash keys, so lookups degrade toward a linear scan → O(n) each, **thousands of times slower**. This is the scenario hash-flooding denial-of-service attacks (sending a server many keys with colliding hashes to make it grind to a halt) exploit, and the reason Java 8 added treeification (and why `String` — being `Comparable` — is well protected as a key).

Sizing a map correctly

`new HashMap<>(100)` does **not** mean "room for 100 entries". The argument is an initial *capacity*, rounded up to a power of two (128); the threshold is then 128 × 0.75 = 96, so the 97th `put` still triggers a resize. To pre-size for an expected number of **entries**, use the Java 19 factories, which do the load-factor maths for you:

```
Map<String, User> users = HashMap.newHashMap(100);       // Java 19: capacity for 100 mappings without resizing
Set<String> seen        = HashSet.newHashSet(100);        // also LinkedHashMap.newLinkedHashMap,
LinkedHashSet.newLinkedHashSet
```

Thread safety

`HashMap` is **not** thread-safe. Concurrent `put`s can lose entries or corrupt a bucket (in Java 7 and earlier, a concurrent resize could even link a chain into a cycle so `get` spun forever). Use `ConcurrentHashMap` (a thread-safe map) for shared maps (Chapter 14). `Collections.synchronizedMap` works but locks the whole map on every call.

Trade-offs

	HashMap	TreeMap
Needs	good `hashCode` + consistent `equals`	`Comparable` keys or a `Comparator`
Average `get` / `put`	O(1)	O(log n)
Worst case	O(log n) per bucket if keys are `Comparable`, else O(n)	O(log n) guaranteed
Order	none	sorted
Memory	table + a node per entry	a tree node per entry (more pointers)

Common mistakes

- Overriding `equals` without `hashCode` (or using different fields in each) → "equal" keys land in different buckets.
- `hashCode` that returns a constant or uses only a low-variety field (e.g. a boolean) → every lookup scans.
- Relying on iteration order: it follows bucket order and can change after a resize or between JDK versions. Use `LinkedHashMap` or `TreeMap` when order matters.

In the interview

30-second answer: "HashMap is an array of buckets. It takes the key's hashCode, spreads the high bits into the low bits, and ANDs with capacity minus one to pick a bucket. Collisions chain in a linked list; since Java 8 a bucket with more than 8 entries becomes a red-black tree if the table has at least 64 buckets, so the worst case is O(log n) instead of O(n). When size passes capacity × 0.75 the table doubles and entries are re-bucketed — each moves to index i or i + oldCapacity. Lookups compare hash first, then equals, which is why equals and hashCode must agree and keys shouldn't be mutable."

- *Follow-up: Why is the capacity a power of two?* → So `hash & (n−1)` replaces a slower modulo, and resizing splits each bucket into exactly two predictable positions.
- *Follow-up: Can HashMap have a null key?* → Yes, one, with hash 0; `Hashtable` and `ConcurrentHashMap` reject nulls.
- *Follow-up: What happens if two keys have the same hashCode but aren't equal?* → Both are stored in the same bucket; lookup distinguishes them with `equals`. Correct but slower.

Self-check

1. Default capacity, load factor and first resize threshold? → 16, 0.75, 12.
2. When does a bucket become a tree? → When adding to a bucket that already holds 8 nodes, provided the table has at least 64 buckets (else it resizes).
3. During a resize from 16 to 32, where can the entry in bucket 5 go? → Bucket 5 or bucket 21.

TL;DR

hashCode → spread → `& (n−1)` → bucket; match by hash then `equals`. Load factor 0.75 triggers doubling; buckets over 8 become red-black trees (table ≥ 64). Good, stable hash codes = O(1); bad ones = slow maps. Pre-size with `HashMap.newHashMap(n)`. Not thread-safe.

10.6 Queues, deques and priority queues

What it is

- A **queue** holds elements waiting to be processed — normally **FIFO** (first in, first out), like a line at a till.
- A **deque** ("deck", double-ended queue) allows adding and removing at both ends, so it can be a FIFO queue *or* a **LIFO stack** (last in, first out, like a stack of plates).
- A **priority queue** always hands out the *smallest* element next (by natural order or a comparator), like a hospital triage desk.

Two method families

`Queue` offers each operation twice: one throws when it can't proceed, the other returns a special value. Prefer the special-value versions for ordinary control flow.

Operation	Throws exception	Returns special value
Insert at tail	`add(e)`	`offer(e)` → `false` if full (bounded queues)
Remove head	`remove()`	`poll()` → `null` if empty
Examine head	`element()`	`peek()` → `null` if empty

`Deque` adds `First`/`Last` versions of each (`offerFirst`, `pollLast`, `peekFirst`…) plus the stack methods `push` (= `addFirst`), `pop` (= `removeFirst`) and `peek`.

How the implementations work

- `ArrayDeque` — a resizable **circular array** with `head` and `tail` indexes that wrap around. Adding/removing at either end is amortised **O(1)** with no node allocations. It rejects `null` (because `null` is how `poll` signals "empty"). The Javadoc notes it is likely faster than `Stack` as a stack and faster than `LinkedList` as a queue. **Use it for both.**
- `PriorityQueue` — a **binary min-heap** stored in an array: a complete binary tree (every level full except possibly the last, which fills left to right) where each parent ≤ its children, and element *i*'s children sit at *2i+1* and *2i+2*. `offer` appends at the end and "sifts up"; `poll` moves the last element to the root and "sifts down". Both are **O(log n)**; `peek` is O(1); `contains`/`remove(Object)` are O(n). Building one from a collection (heapify) is O(n). The array is *not* sorted — only the root is guaranteed smallest — so `toString()` and the iterator show heap order, not sorted order.

Code

```
// code/ch10/queues/QueueDemo.java (key parts)
Queue<String> printer = new ArrayDeque<>();
printer.offer("doc1"); printer.offer("doc2"); printer.offer("doc3");

Deque<Character> stack = new ArrayDeque<>();                // balanced-brackets check
for (char c : "{[()]}(".toCharArray()) {
    if ("([{".indexOf(c) >= 0) stack.push(c);
    else if (!stack.isEmpty() && "([{".indexOf(stack.peek()) == ")]}".indexOf(c)) stack.pop();
    else { stack.push('!'); break; }
}

PriorityQueue<Integer> heap = new PriorityQueue<>(List.of(7, 3, 9, 1, 5));
while (!heap.isEmpty()) out.append(heap.poll()).append(' ');

PriorityQueue<Task> jobs = new PriorityQueue<>(
        Comparator.comparingInt(Task::priority).reversed().thenComparing(Task::name));

int k = 3; PriorityQueue<Integer> topK = new PriorityQueue<>();  // top-k largest with a min-heap
for (int x : new int[]{15, 2, 99, 42, 8, 77, 23}) { topK.offer(x); if (topK.size() > k) topK.poll(); }
```

Output:

```
peek=doc1 poll=doc1 left=[doc2, doc3]
empty.poll()=null empty.peek()=null
empty.remove() -> NoSuchElementException
{[()]}( balanced? false (left on stack: [(])
ArrayDeque.push(null) -> NPE
heap toString (NOT sorted): [1, 3, 9, 7, 5]
poll order: 1 3 5 7 9
backup deploy lint email
top-3 = [99, 77, 42]
```

The top-k trick keeps a heap of only *k* elements: each new element costs O(log k), and the smallest of the current top k is evicted. Total O(n log k) instead of sorting everything in O(n log n) — a staple of coding interviews (Chapter 19).

When not to use them

- Threads producing and consuming (the producer–consumer pattern: some threads add work, others take it) → a `BlockingQueue` (`ArrayBlockingQueue`, `LinkedBlockingQueue`) that waits instead of returning `null` (Chapter 14).
- You need all elements in sorted order once → just sort a list; a heap shines when elements keep arriving.
- Changing an element's priority after insertion is not supported — remove and re-add it (O(n)), or insert a new entry and skip stale ones when polled.

Common mistakes

- Using `Stack` or `LinkedList` where `ArrayDeque` fits.
- Printing a `PriorityQueue` and expecting sorted output; iterate with `poll()` instead.
- Mixing `add`/`remove` (exceptions) with `offer`/`poll` expectations.
- Using `push` then `pollLast` — `push` adds at the **front** of a deque.

Self-check

1. Which call returns `null` rather than throwing on an empty queue? → `poll()` / `peek()`.
2. What does `PriorityQueue.poll()` return? → The smallest element per its ordering.
3. Why can't `ArrayDeque` hold `null`? → `null` is the "empty" return value of `poll`/`peek`.

TL;DR

`ArrayDeque` for FIFO queues and LIFO stacks; `PriorityQueue` (binary heap) when the next item is "the smallest": O(log n) add/poll, O(1) peek, unsorted iteration.

10.7 Iterators and fail-fast behaviour

What it is

An `Iterator<E>` is a cursor over a collection with three methods: `hasNext()`, `next()` and the optional `remove()`. The enhanced for loop is compiler shorthand for using one. `ListIterator` adds `hasPrevious`/`previous`, index queries, `set` and `add`.

How fail-fast works

Most `java.util` collections keep a **modification counter** (`modCount`) that increments on every structural change. An iterator records the count when created and re-checks it on each `next()`. If the collection was changed by anything *other than the iterator itself*, `next()` throws `ConcurrentModificationException` (CME). Despite the name, this happens in single-threaded code too. It is "best effort": it catches bugs early but is not a thread-safety guarantee.

Fail-safe iterators (in `java.util.concurrent`) iterate over a snapshot (a frozen copy) or a weakly consistent view (one that may or may not show concurrent changes, but never fails) and never throw CME.

`CopyOnWriteArrayList` copies the whole array on every write, so it suits read-mostly lists like event listeners (objects notified when something happens).

Code

```
// code/ch10/iteration/IteratorDemo.java (key parts)
for (Integer n : nums) if (n % 2 == 0) nums.remove(n);          // WRONG: CME

for (Iterator<Integer> it = nums.iterator(); it.hasNext(); )    // RIGHT: iterator's own remove
    if (it.next() % 2 == 0) it.remove();

nums.removeIf(n -> n % 2 == 0);                                 // SIMPLEST (Java 8)

for (ListIterator<String> li = words.listIterator(); li.hasNext(); ) {
    String w = li.next();
    li.set(w.toUpperCase());
    if (w.equals("b")) li.add("b2");                            // inserted after the cursor
}
scores.entrySet().removeIf(e -> e.getValue() < 50);             // remove map entries by condition

List<String> listeners = new CopyOnWriteArrayList<>(List.of("L1", "L2"));
for (String l : listeners) listeners.add(l + "'");              // iterates the old snapshot
```

Output:

```
for-each + list.remove -> ConcurrentModificationException, list now [1, 3, 4, 5, 6]
iterator.remove -> [1, 3, 5]
removeIf        -> [1, 3, 5]
listIterator set/add -> [A, B, b2, C]
backwards -> Cb2BA
passed -> {bo=90, cy=55}
CopyOnWriteArrayList -> [L1, L2, L1', L2']
```

The first line shows the list was already changed (`2` removed) before the exception — fail-fast stops you *after* the first bad change, so don't rely on catching CME.

Common mistakes

- Removing inside for-each (above). Also: calling `it.remove()` twice per `next()`, or before the first `next()` → `IllegalStateException`.
- Believing CME means "a threading problem" — usually it is a loop modifying its own collection.
- Removing via index in a forward `for (int i…)` loop and skipping the element that shifted into position `i`; loop backwards or use `removeIf`.

Self-check

1. Name two safe ways to remove while iterating. → `Iterator.remove()` and `Collection.removeIf`.
2. Does CME need two threads? → No.
3. Why doesn't `CopyOnWriteArrayList` throw CME? → Its iterator walks an immutable snapshot of the array.

TL;DR

Don't structurally modify a collection while for-each-ing over it. Use `removeIf`, `Iterator.remove`, or collect changes and apply them after. Fail-fast = bug detector, not a lock.

10.8 Ordering: `Comparable` and `Comparator`

What it is

Sorting needs a rule for "which comes first".

- `Comparable<T>` — the class defines its **natural order** by implementing `int compareTo(T other)`. `String`, `Integer`, `LocalDate` and enums (declaration order) are `Comparable`.

- `Comparator<T>` — a separate object with `int compare(T a, T b)` that defines *any* order you want, without changing the class. Analogy: `Comparable` is a book's own shelf number; a `Comparator` is you choosing to arrange books by colour today.

Both return a **negative** number if the first comes before the second, **zero** if they tie, and **positive** if it comes after. Only the sign matters.

The contract

`compareTo` / `compare` must be **antisymmetric** (sign of `a.compareTo(b)` is the opposite of `b.compareTo(a)`), **transitive** (if a comes before b and b before c, then a comes before c), and ideally **consistent with equals** (`compareTo == 0` exactly when `equals` is true). Sorted collections (`TreeSet`, `TreeMap`) use the comparison *instead of* `equals`, so an inconsistent comparator makes them drop or merge elements. (`BigDecimal` is a famous JDK exception: `2.0` and `2.00` compare equal but aren't `equals`.)

Building comparators (Java 8+)

Factory / method	Meaning
`Comparator.comparing(Employee::name)`	by a `Comparable` key
`comparingInt/Long/Double(keyFn)`	by a primitive key (no boxing)
`comparing(keyFn, keyComparator)`	by a key using a given comparator
`.thenComparing(...)`	tie-breaker
`.reversed()`	flip the whole comparator so far
`Comparator.naturalOrder()` / `reverseOrder()`	natural / reverse natural
`nullsFirst(c)` / `nullsLast(c)`	tolerate nulls
`String.CASE_INSENSITIVE_ORDER`	ready-made

Code

```java
// code/ch10/sorting/SortingDemo.java (key parts)
record Employee(int id, String name, String dept, double salary) implements Comparable<Employee> {
    @Override public int compareTo(Employee o) { return Integer.compare(id, o.id); }   // natural order: id
}
Collections.sort(staff);                                              // uses compareTo
staff.sort(Comparator.comparing(Employee::name).reversed());
staff.sort(Comparator.comparing(Employee::dept)
        .thenComparing(Employee::salary, Comparator.reverseOrder())   // reverse just this key
        .thenComparing(Employee::name));
staff.sort(Comparator.comparingDouble(Employee::salary));
cities.sort(Comparator.nullsLast(String.CASE_INSENSITIVE_ORDER));

Comparator<Integer> bad = (a, b) -> a - b;                            // overflow bug
Set<String> byLength = new TreeSet<>(Comparator.comparingInt(String::length));
byLength.addAll(List.of("kiwi", "pear", "fig", "plum", "banana"));
```

Output:

```
by id:      [Ana, Bo, Cy, Dee]
name desc:  [Dee, Cy, Bo, Ana]
dept,salary desc,name: [Bo, Cy, Dee, Ana]
salary asc: [Ana, Cy, Dee, Bo]
nullsLast, case-insensitive: [Accra, lima, Oslo, null]
bad.compare(MIN_VALUE, 1) = 2147483647 (positive, so MIN_VALUE looks BIGGER), Integer.compare = -1
TreeSet by length keeps only [fig, kiwi, banana]  (pear and plum were dropped!)
stable (dept, then previous name order): [Bo, Cy, Dee, Ana]
```

- `reversed()` applies to **everything chained before it**; to reverse only one key, pass `Comparator.reverseOrder()` as that key's comparator (as done for salary).

- `a - b` overflows for large magnitudes (`MIN_VALUE − 1` wraps to `MAX_VALUE`). Always use `Integer.compare`, `Long.compare`, `Double.compare` or `comparingInt`.
- The `TreeSet` ordered by length considers `"kiwi"`, `"pear"` and `"plum"` duplicates (length 4 ties), keeping only the first. Add a tie-breaker: `comparingInt(String::length).thenComparing(Comparator.naturalOrder())`.
- **Stability.** `List.sort` / `Collections.sort` (TimSort — a merge-sort-based algorithm — O(n log n)) are **stable**: equal elements keep their relative order. That's why sorting by name, then by dept, leaves each dept sorted by name.

Common mistakes

- Subtraction comparators; comparing `double` fields with `<` / `>` and forgetting `NaN` (use `Double.compare`).
- `Comparable` whose natural order disagrees with `equals` used as `TreeMap` keys.
- Putting non-`Comparable` objects into a `TreeSet` without a comparator → `ClassCastException` on the first add.

In the interview

> **30-second answer:** "Comparable is the class's single natural order via compareTo; Comparator is an external strategy (a plug-in object holding one way of doing something) for any number of orders. Since Java 8 I build comparators with comparing, thenComparing and reversed, use primitive variants to avoid boxing, and never subtract to compare. Sorted collections treat compare == 0 as equality, so the order should be consistent with equals."

- *Follow-up: Is `Collections.sort` stable?* → Yes (TimSort for objects).
- *Follow-up: How do you sort descending by one key but ascending by another?* → `comparing(a).thenComparing(b, reverseOrder())`.

Self-check

1. What does a negative return from `compare(a, b)` mean? → `a` sorts before `b`.
2. How to sort `List<String>` by length, then alphabetically? → `list.sort(comparingInt(String::length).thenComparing(naturalOrder()))`.
3. Why is `(a, b) -> a - b` dangerous? → Integer overflow can flip the sign.

TL;DR

`Comparable` = one built-in order; `Comparator` = any order, composable with `comparing` / `thenComparing` / `reversed` / `nullsLast`. Keep it consistent with `equals` for sorted collections. Sorting objects is stable.

10.9 The `Collections` utility class

`java.util.Collections` (plural, with an *s*) holds static algorithms and wrappers — the collection counterpart of `java.util.Arrays` (§4.8).

```java
// code/ch10/utilities/CollectionsUtilDemo.java (key parts)
Collections.max(xs); Collections.min(xs); Collections.frequency(xs, 3);
Collections.sort(xs); Collections.binarySearch(xs, 8);    // binarySearch needs a sorted list
Collections.reverse(xs); Collections.swap(xs, 0, xs.size() - 1);
Collections.rotate(xs, 2); Collections.shuffle(xs, new Random(42));   // seeded -> repeatable
Collections.nCopies(3, "ab"); Collections.disjoint(List.of(1, 2), List.of(3));
Collections.addAll(letters, "d", "b");

List<String> view = Collections.unmodifiableList(source);  // read-only VIEW
List<String> copy = List.copyOf(source);                   // immutable COPY
source.add("y");

List<Integer> sync = Collections.synchronizedList(new ArrayList<>(List.of(1, 2)));
synchronized (sync) { for (int v : sync) System.out.print("sync " + v + " "); }   // lock while iterating
```

Output:

```
max=9 min=1 frequency(3)=2
sorted=[1, 3, 3, 5, 8, 9] binarySearch(8)=4 binarySearch(4)=-4 (negative: -(insertionPoint)-1)
reverse=[9, 8, 5, 3, 3, 1]
swap ends=[1, 8, 5, 3, 3, 9]
rotate 2=[3, 9, 1, 8, 5, 3]
shuffle(seed 42)=[9, 3, 5, 3, 8, 1]
nCopies=[ab, ab, ab] disjoint=true
addAll varargs=[c, a, d, b]
view sees change: [x, y]  copy does not: [x]
view.add -> UnsupportedOperationException
emptyList=[] singleton=[only]
sync 1 sync 2
```

Group	Methods
Algorithms	`sort`, `binarySearch`, `reverse`, `shuffle`, `swap`, `rotate`, `fill`, `copy`, `min`, `max`, `frequency`, `disjoint`, `addAll`, `nCopies`
Read-only wrappers (views)	`unmodifiableList/Set/Map/Collection`, `unmodifiableSequencedCollection/Set/Map` (21), sorted/navigable variants
Synchronized wrappers	`synchronizedList/Set/Map/...` — each call locks; you must lock manually around iteration
Constants and singletons	`emptyList/Set/Map`, `singletonList/singleton/singletonMap` (immutable)
Type-checked wrappers	`checkedList(list, String.class)` — catches heap pollution (a generic collection ending up with the wrong type inside) from raw types (generics used without `<…>`) at insertion time (Chapter 9)
Other	`reverseOrder()`, `newSetFromMap(map)`, `list(enumeration)`, `enumeration(coll)`

`binarySearch` returns the index if found, otherwise `-(insertionPoint) - 1` — always negative, so "not found" can't be confused with index 0. Here 4 would be inserted at index 3, so the result is −4.

Self-check

1. `Collections.binarySearch` on an unsorted list? → Undefined result — sort first.
2. What's the difference between `unmodifiableList(l)` and `List.copyOf(l)`? → The first is a read-only view that reflects later changes to `l`; the second is an independent immutable snapshot.

10.10 Unmodifiable and immutable collections

What it is

- **Unmodifiable** — the collection object rejects changes (`add`, `set`, `remove` throw `UnsupportedOperationException`).
- **Immutable** — it is unmodifiable *and* nothing else can change it either (no backing collection behind it).
- A **view** is a wrapper over another collection; it can be unmodifiable yet still change when the underlying collection changes.

Why it matters

Returning an immutable collection from a method (or storing one in a record) means callers can't corrupt your object's state, the collection is automatically safe to share across threads, and intent is explicit. *Effective Java* (Joshua Bloch's best-practice book) Item 17 ("minimise mutability") and Item 50 ("make defensive copies when needed" — private copies so outsiders can't change your data) both point here.

The factory methods

Factory	Since	Nulls	Duplicates	Order
`List.of(...)`	9	NPE	allowed	as given
`Set.of(...)`	9	NPE	**`IllegalArgumentException`**	unspecified (randomised per JVM run)
`Map.of(k1, v1, … up to 10 pairs)` / `Map.ofEntries(Map.entry(k, v), …)`	9	NPE	duplicate key → IAE	unspecified
`List.copyOf` / `Set.copyOf` / `Map.copyOf`	10	NPE	`Set.copyOf` drops duplicates	—
`Stream.toList()`	16	allowed	allowed	encounter order
`Collectors.toUnmodifiableList/Set/Map`	10	NPE	—	—
`Arrays.asList(...)`	1.2	allowed	allowed	fixed-size, **`set` works**, write-through to the array
`Collections.unmodifiableX(c)`	1.2	as `c`	as `c`	read-only **view** of `c`

Code

```
// code/ch10/immutable/ImmutableDemo.java (key parts)
List<String> days = List.of("Mon", "Tue", "Wed");
attempt("List.of(...).add", () -> days.add("Thu"));
attempt("List.of with null", () -> List.of("a", null));
attempt("List.of(...).contains(null)", () -> days.contains(null));
attempt("Set.of duplicate", () -> Set.of("a", "a"));
attempt("Map.of duplicate key", () -> Map.of("k", 1, "k", 2));
List<String> snap = List.copyOf(src); src.add("b");
List<Integer> t = Stream.of(3, 1, 2).toList();
List<StringBuilder> sbs = List.of(new StringBuilder("hi")); sbs.get(0).append("!");
```

Output:

```
List.of(...).add -> UnsupportedOperationException
List.of(...).set -> UnsupportedOperationException
List.of with null -> NullPointerException
List.of(...).contains(null) -> NullPointerException
Set.of duplicate -> IllegalArgumentException
Map.of duplicate key -> IllegalArgumentException
Map.ofEntries get = 404
Arrays.asList set -> ok
Arrays.asList add -> UnsupportedOperationException
copyOf snapshot=[a] src=[a, b] copyOf(List.of) same object? true
Stream.toList().add -> UnsupportedOperationException
element mutated inside an 'immutable' list: [hi!]
Set.of size=3 (print order may differ between runs)
```

Points to remember:

- These collections are **shallowly** immutable: the list can't change, but mutable elements inside can. Store immutable elements (strings, records of immutable fields) for true immutability.
- `List.of(...).contains(null)` throws `NullPointerException` — a surprise when code passes a nullable value to `contains`.
- `copyOf` is free when its argument is already an unmodifiable `List.of`-style list (it returns the same object).
- `Collectors.toList()` makes no promise about mutability (today it returns an `ArrayList`); if you want an unmodifiable result, say so with `Stream.toList()` (Chapter 12).

Defensive copy in a record — records are shallowly immutable (§6.8), so copy collection components in the compact constructor:

```
record Team(String name, List<String> members) {
    Team { members = List.copyOf(members); }   // callers can't change our list later, and we reject nulls
}
```

Self-check

1. Which factory throws on a duplicate element? → `Set.of` (and `Map.of` on a duplicate key).
2. Can you `set(0, x)` on `Arrays.asList(...)`? → Yes; you can't `add`/`remove`.
3. Is `List.of(new StringBuilder())` deeply immutable? → No — the builder inside is still mutable.

TL;DR

Prefer `List.of`/`Set.of`/`Map.of` for constants and `List.copyOf` for defensive copies; they reject nulls and are shallowly immutable. `unmodifiableList` is only a view. `Arrays.asList` is fixed-size but writable.

10.11 Sequenced collections (Java 21)

The problem it solves

Before Java 21 "get the first/last element" or "iterate in reverse" had a different spelling for every type: `list.get(0)` and `list.get(list.size() - 1)`, `deque.getFirst()`, `sortedSet.first()`/`last()`, and for a `LinkedHashSet` no direct way to get the last element at all. There was also no common type meaning "a collection with a defined order".

What it is

JEP 431 (a JEP is a numbered JDK Enhancement Proposal; final in Java 21) added three interfaces that capture "has a well-defined encounter order":

Interface	New methods	Retrofitted onto
`SequencedCollection<E>`	`addFirst`, `addLast`, `getFirst`, `getLast`, `removeFirst`, `removeLast`, `reversed()`	`List`, `Deque`
`SequencedSet<E>`	`reversed()` returns a `SequencedSet`	`LinkedHashSet`, `SortedSet` (so `TreeSet`)
`SequencedMap<K,V>`	`firstEntry`, `lastEntry`, `pollFirstEntry`, `pollLastEntry`, `putFirst`, `putLast`, `reversed()`, `sequencedKeySet()`, `sequencedValues()`, `sequencedEntrySet()`	`LinkedHashMap`, `SortedMap` (so `TreeMap`)

Plus `Collections.unmodifiableSequencedCollection/Set/Map`. `HashSet` and `HashMap` are **not** sequenced (they have no defined order).

How it behaves

1. `reversed()` returns a **live reverse-ordered view**, not a copy — O(1) to create.
2. `getFirst`/`removeFirst` etc. on an empty collection throw `NoSuchElementException`.
3. For sorted collections (`TreeSet`, `TreeMap`), `addFirst`/`addLast`/`putFirst`/`putLast` throw `UnsupportedOperationException`: position is decided by the comparator, not by you.
4. For `LinkedHashSet`/`LinkedHashMap`, `addFirst`/`putFirst` on an element that's already present **moves** it to the front.
5. Unmodifiable collections throw `UnsupportedOperationException` from the mutating methods, as usual.

Code

```
// code/ch10/sequenced/SequencedDemo.java (key parts)
List<String> list = new ArrayList<>(List.of("b", "c"));
list.addFirst("a"); list.addLast("d");
list.getFirst(); list.getLast(); list.reversed();

LinkedHashSet<String> tags = new LinkedHashSet<>(List.of("java", "sql", "git"));
tags.addFirst("git");                  // already present -> moved to the front

TreeSet<Integer> sorted = new TreeSet<>(List.of(30, 10, 20));
sorted.addFirst(5);                    // UnsupportedOperationException

List<String> rev = list.reversed(); list.add("e");   // the view sees the change

LinkedHashMap<String, Integer> steps = new LinkedHashMap<>();
steps.put("build", 1); steps.put("test", 2); steps.put("deploy", 3);
steps.putFirst("checkout", 0);
steps.firstEntry(); steps.lastEntry(); steps.reversed().keySet(); steps.pollFirstEntry();

static <E> E lastOf(SequencedCollection<E> c) { return c.getLast(); }   // one method for any ordered collection
```

Output:

```
List        first=a last=d reversed=[d, c, b, a]
LinkedHashSet [git, java, sql] removeLast=sql -> [git, java]
Deque       reversed=[3, 2, 1] getLast=3
TreeSet     first=10 last=30 reversed=[30, 20, 10]
TreeSet.addFirst -> UnsupportedOperationException (order comes from the comparator)
view after list.add(e): [e, d, c, b, a]
SequencedMap first=checkout=0 last=deploy=3 reversed keys=[deploy, test, build, checkout]
pollFirstEntry=checkout=0 -> [build, test, deploy]
lastOf(list)=e lastOf(dq)=3 lastOf(tags)=java
empty getFirst -> NoSuchElementException
HashSet is sequenced? false
```

Common mistakes

- Assuming `reversed()` copies — later changes to the original show through (and vice versa for mutable views).
- Calling `addFirst` on a `TreeSet` or on `List.of(...)`.
- Compatibility note: a home-grown class that implements `List` or `Deque` and already declared a method such as `reversed()` or `getFirst()` with a different return type stops compiling on Java 21. JEP 431 lists this as a known source-compatibility risk.

Self-check

1. Which of `HashSet`, `LinkedHashSet`, `TreeSet` are `SequencedSet`s? → `LinkedHashSet` and `TreeSet`.
2. What does `list.reversed()` cost? → O(1); it's a view.
3. What happens on `new LinkedHashMap<>().firstEntry()`? → Returns `null` for an empty map (the `SequencedMap` entry methods return `null`; only `getFirst`/`removeFirst`-style collection methods throw).

TL;DR

Java 21 gives every ordered collection the same `getFirst/getLast/addFirst/addLast/removeFirst/removeLast/reversed` API (and maps `firstEntry/lastEntry/pollFirstEntry/putFirst/...`). `reversed()` is a view; sorted collections refuse positional adds.

10.12 Big-O cheat sheet and a measured demo

Big-O describes how an operation's cost grows with the number of elements *n*: O(1) = constant, O(log n) = grows very slowly (tree height), O(n) = proportional (scan). "Amortised" means averaged over many calls; "average" for hash structures assumes a reasonable `hashCode`.

Collection	add (end)	add/remove front	get(i)	contains / get(key)	remove(element/key)	Iteration order	Nulls
`ArrayList`	O(1)*	O(n)	O(1)	O(n)	O(n)	index	yes
`LinkedList`	O(1)	O(1)	O(n)	O(n)	O(n)	index	yes
`ArrayDeque`	O(1)*	O(1)*	—	O(n)	O(n)	head→tail	**no**
`PriorityQueue`	O(log n)	poll O(log n)	—	O(n)	O(n)	heap (unsorted)	no
`HashSet` / `HashMap`	O(1)†	—	—	O(1)†	O(1)†	none	one null key
`LinkedHashSet` / `LinkedHashMap`	O(1)†	—	—	O(1)†	O(1)†	insertion/access	one null key
`TreeSet` / `TreeMap`	O(log n)	—	—	O(log n)	O(log n)	sorted	no (natural order)
`EnumSet` / `EnumMap`	O(1)	—	—	O(1)	O(1)	enum order	no
`CopyOnWriteArrayList`	O(n)	O(n)	O(1)	O(n)	O(n)	index	yes

* amortised. † average; worst case O(log n) for a treeified bucket (or O(n) for non-`Comparable` keys with colliding hashes).

Measured demo

`CollectionBenchmark` times a few operations once after a warm-up run (running the code first so the JIT compiler has optimised it). It is a *demonstration*, not a rigorous benchmark — use JMH (Java

Microbenchmark Harness) for real measurements (Chapter 16). **Timings are machine-dependent** (this run: Apple M4 Pro, JDK 26); what matters is the ratio within each row.

```
// code/ch10/perf/CollectionBenchmark.java (key parts)
Runnable getA = () -> { for (int i = 0; i < 20_000; i++) sink += array.get((i * 7919) % n); };
Runnable getL = () -> { for (int i = 0; i < 20_000; i++) sink += linked.get((i * 7919) % n); };
Runnable frontA = () -> { List<Integer> l = new ArrayList<>(); for (int i = 0; i < 50_000; i++) l.add(0, i); };
Runnable frontD = () -> { Deque<Integer> d = new ArrayDeque<>(); for (int i = 0; i < 50_000; i++) d.addFirst(i);
};
Runnable cL = () -> { for (int i = 0; i < 2_000; i++) sink += array.contains(n - 1 - i) ? 1 : 0; };
Runnable cH = () -> { for (int i = 0; i < 2_000; i++) sink += hash.contains(n - 1 - i) ? 1 : 0; };
// ... TreeSet contains, full iteration, presized vs default ArrayList
```

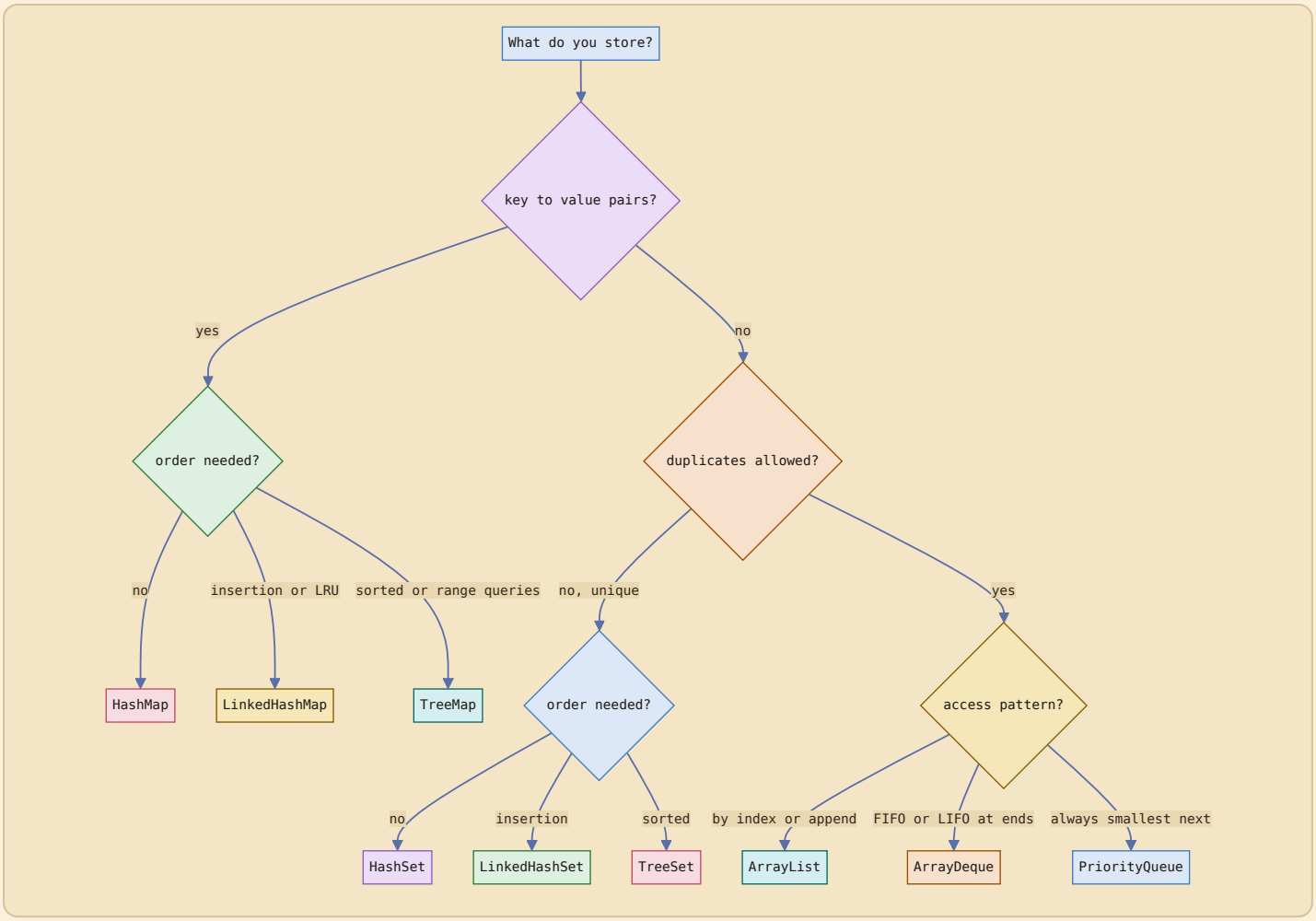
Output:

```
20k get(i) n=100k        ArrayList     0.6 ms | LinkedList   870.1 ms
50k add-at-front          ArrayList    83.6 ms | ArrayDeque     1.0 ms
2k contains n=100k         ArrayList    65.7 ms | HashSet 0.1 ms | TreeSet 0.3 ms
20 full iterations       ArrayList     4.2 ms | LinkedList     4.1 ms
2M adds                   default    42.7 ms | presized     7.7 ms
```

What each row shows:

1. **Indexing** — `LinkedList.get(i)` walks up to 50,000 nodes per call: >1000× slower.
2. **Front inserts** — `ArrayList.add(0, x)` shifts everything (O(n²) total); `ArrayDeque.addFirst` is O(1).
3. **Membership** — list scan vs hashing vs a ~17-level tree walk.
4. **Iteration** — both O(n); these freshly allocated nodes happen to sit close together, so the gap is small. In long-running apps with fragmented heaps (free memory scattered in many small gaps), linked structures usually iterate noticeably slower.
5. **Pre-sizing** — skipping ~30 grow-and-copy steps (and their garbage — discarded arrays the garbage collector must clean up) is several times faster.

10.13 Choosing a collection



Extra rules of thumb:

- Enum keys or elements → `EnumMap` / `EnumSet`.
- Fixed, known-at-compile-time data → `List.of` / `Set.of` / `Map.of`.
- Shared between threads → `ConcurrentHashMap`, `CopyOnWriteArrayList` (read-mostly), `BlockingQueue` (Chapter 14).
- Millions of primitives in a hot path → primitive arrays (or a primitive-collections library such as Eclipse Collections or fastutil) to avoid boxing.
- In doubt: `ArrayList` and `HashMap`, then measure.

Python readers: the same ideas as `list`, `set`, `dict`, `deque` and `heapq` in Python Chapter 6.

In the interview — collections quick round

- *`HashSet` vs `TreeSet`?* → O(1) unordered vs O(log n) sorted with range queries; `TreeSet` needs `Comparable` / `Comparator`, uses compare-to-zero for equality.
- *Why is `ArrayDeque` preferred over `Stack`?* → `Stack` extends synchronized `Vector` (locking overhead, exposes index methods); `ArrayDeque` is a lean circular array.
- *How would you build an LRU cache?* → `LinkedHashMap` with `accessOrder = true` and `removeEldestEntry` (or `HashMap` + doubly linked list by hand — LLD case studies).
- *Is `List.of` thread-safe?* → Yes for reading — it's immutable; the elements themselves may not be.

Chapter TL;DR

Topic	Remember
Hierarchy	`Iterable` → `Collection` → `List` / `Set` / `Queue` / `Deque`; `Map` separate with `keySet` / `values` / `entrySet` views; program to interfaces
`ArrayList`	growable array (10, then ×1.5): O(1) get/append, O(n) middle edits; the default list
`LinkedList`	doubly linked nodes: O(1) ends, O(n) get; rarely the best choice
Sets	`HashSet` O(1), `LinkedHashSet` insertion order, `TreeSet` sorted O(log n) + `floor` / `ceiling`; never mutate elements
Map API	`getOrDefault`, `putIfAbsent`, `merge`, `compute*`; iterate `entrySet`; `TreeMap` for ranges; `LinkedHashMap` for LRU
`HashMap` internals	hash spread → `& (n−1)`; chain then tree (>8 in bucket, table ≥ 64); load factor 0.75 → double; equal hash then `equals`; `newHashMap(n)` (19); not thread-safe
Queues	`ArrayDeque` for queue and stack; `PriorityQueue` = binary heap (O(log n), unsorted iteration); `offer/poll/peek` return special values
Iteration	fail-fast CME; remove with `removeIf` or `Iterator.remove`
Ordering	`Comparable` natural vs `Comparator` composed; never subtract; consistent with equals for `TreeSet` / `TreeMap`; sorts are stable
Immutability	`List/Set/Map.of` (9), `copyOf` (10): no nulls, shallow; `Stream.toList` (16) unmodifiable but allows nulls; `unmodifiableX` = view; `Arrays.asList` fixed-size
Sequenced (21)	`getFirst/getLast/addFirst/addLast/removeFirst/removeLast/reversed` on lists, deques, `LinkedHashSet`, sorted sets; maps get `firstEntry/pollFirstEntry/putFirst/...`

Where next: pass behaviour into collections methods cleanly with lambdas and method references in Chapter 11, then transform collections declaratively with Chapter 12 — Streams. Exam traps: OCP Chapter 9. Thread-safe collections: Chapter 14.